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Shape stability of a gas bubble in a soft solid

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ABSTRACT

Predicting the onset of non-spherical oscillations of bubbles in soft matter is a fundamental cavitation problem with implications to sonoprocessing, polymeric materials synthesis, and biomedical ultrasound applications. The shape stability of a bubble in a Kelvin-Voigt viscoelastic medium with nonlinear elasticity, the simplest constitutive model for soft solids, is analytically investigated and compared to experiments. Using perturbation methods, we develop a model reducing the equations of motion to two sets of evolution equations: a Rayleigh-Plesset-type equation for the mean (volume-equivalent) bubble radius and an equation for the non-spherical mode amplitudes. Parametric instability is predicted by examining the natural frequency and the Mathieu equation for the non-spherical modes, which are obtained from our model. Our theoretical results show good agreement with published experiments of the shape oscillations of a bubble in a gelatin gel. We further examine the impact of viscoelasticity on the time evolution of non-spherical mode amplitudes. In particular, we find that viscosity increases the damping rate, thus suppressing the shape instability, while shear modulus increases the natural frequency, which changes the unstable mode. We also explain the contributions of rotational and irrotational fields to the viscoelastic stresses in the surroundings and at the bubble surface, as these contributions affect the damping rate and the unstable mode. Our analysis on the role of viscoelasticity is potentially useful to measure viscoelastic properties of soft materials by experimentally observing the shape oscillations of a bubble.

1. Introduction

The oscillations of bubbles driven by external sound fields can lead to important physical, chemical, and biological outcomes. Rapid compression of bubbles gives rise to elevated temperatures and pressures, which can induce specific chemical reactions (sonochemistry) [1–3] and generate shock waves capable of damaging surrounding tissue in therapeutic ultrasound [4–6]. Repeated oscillations also induce microstreaming and local stress fields in biomedical and cleaning applications [7–9]. Acoustic cavitation and its outcomes play important roles in sonoprocessing [10], materials synthesis [11], and biomedical applications including drug delivery [12–14]. In water treatment processes, chemical pollutants can be degraded through ultrasound-induced cavitation [10]. Novel materials such as polymers with improved surface functionality can be developed using sonochemistry at solid surfaces [11]. The shock waves generated by the bubble collapse can enhance skin permeability for transdermal drug delivery (sonophoresis) [12]. In targeted drug delivery, microbubbles are injected in the vasculature and flow to pre-identified regions where focused ultrasound induces bubble break up thereby releasing drugs [13,14].

To date, most of the analysis enabling the prediction of these outcomes of cavitation has relied on spherical bubble dynamics. However, the bubble dynamics may depart from a spherical behavior, as is the case in cavitation near an interface separating different media [15,4,16] or in parametric instabilities of continuously forced bubbles [17,18]. Outcomes include non-spherical oscillations, which can lead to more effective gas mixing in sonochemistry [3], mitigation of temperature rise to change the efficiency of sonochemical production [19,2], or the formation of high-speed liquid jets capable of causing structural damage [15,4,16]. Furthermore, the role of viscoelastic properties on such behaviors is poorly understood.

Stability analysis of a gas bubble in water was pioneered by Plesset [20] and Prosperetti [21] using perturbation methods. Plesset [20] applied spherical harmonics to express the non-spherical bubble surface, and reduced the Bernoulli equation to two ordinary differential equations (ODEs): the Rayleigh-Plesset equation for the mean bubble radius and an equation for non-spherical mode amplitudes. Prosperetti [21] extended this inviscid model to account for rotational flow by adding two corrections: a viscous correction in the dynamic boundary condition and a rotational correction in the velocity field. Based on this model, the shape stability of a bubble was analytically investigated, primarily motivated by sonoluminescence [22–25], drug delivery

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[26,27] and sonochemistry [28–30]. Recent progress in experimental techniques have enabled direct observations of shape oscillations of a bubble in water [31,17] and in gelatin gels [18] under continuous forcing. Versluis et al. [17] observed parametric instability of non-spherical modes and validated the stability analysis in detail. However, most studies focused on the stability analysis of a bubble in water. A few studies extended stability analysis to a bubble in Maxwell-type viscoelastic fluids [32] or purely elastic media [33], but it has yet to be extended to a bubble in a viscoelastic solid.

Accounting for the viscoelastic nature of the surrounding soft solid is a challenge when modeling bubble dynamics. In the past, different constitutive models have been incorporated into the Rayleigh-Plesset equation, e.g., Maxwell [34,35], Kelvin-Voigt [36,37], and Zener [38]. Estrada et al. [39] showed good agreement between laser-induced inertial bubble collapse in polyacrylamide gels and a Kelvin-Voigt model with nonlinear elasticity. Hamaguchi and Ando [18] experimentally observed spherical and non-spherical bubble oscillations in gelatin gels induced by an ultrasound wave, where the spherical oscillations were described by the Rayleigh-Plesset equation coupled with a Kelvin-Voigt model. These validation studies suggest that a Kelvin-Voigt model appropriately represents bubble dynamics in soft materials.

The goal of this study is to extend shape-stability analysis [20,21] to a bubble in a Kelvin-Voigt viscoelastic medium. The natural frequency of the non-spherical mode is obtained from our model, which can be used to predict the unstable mode due to parametric instability described by a Mathieu equation. We also discuss the validity of our model against the experiments of Hamaguchi and Ando [18], in which shape oscillations of a bubble in a gelatin gel were observed. Lastly, we investigate the shape stability of a bubble under pulse-wave ultrasound irradiation and examine viscoelastic effects on non-spherical mode amplitudes.

2. Physical model

2.1. General formulation

We consider a non-spherical bubble in a homogeneous viscoelastic medium described in spherical coordinates by the radial distance r, the polar angle θ and the azimuthal angle φ, as shown in Fig. 1, where R is the mean (volume-equivalent) bubble radius, $\mathbf{u}$ is the velocity vector, p_b is the bubble pressure, t is time, and ρ, μ and G are the density, viscosity and shear modulus of the surrounding medium. The interface between the gas and the surroundings is infinitesimally thin, with no elastic membrane. The bubble surface can be expressed by the superposition of spherical harmonics,

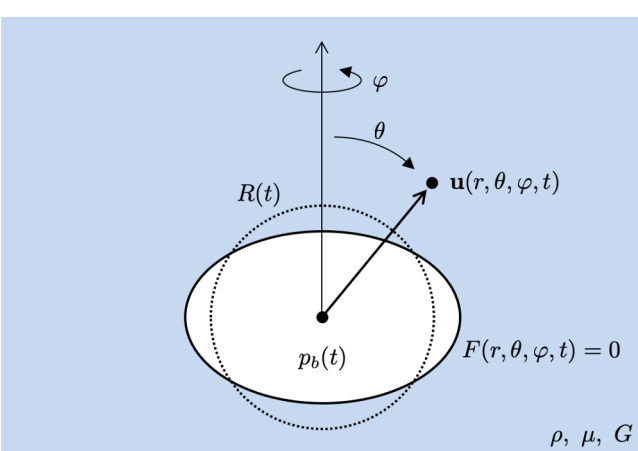

Fig. 1. Non-spherical bubble geometry in spherical coordinates.

$$F(r,\theta,\varphi,t) = r - R(t) - \sum_{n,m} a_n(t) Y_{nm}(\theta,\varphi) = 0, \tag{1}$$

where a_n is the perturbation amplitude of the spherical harmonic mode Y_{nm} of degree n and orders m ($|m| \leqslant n$).

The dynamics of a viscoelastic medium are described by Cauchy's equations of motion,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = \frac{1}{\rho}\mathrm{div}\mathbf{T}, \tag{2}$$

where $\mathbf{T}$ is the Cauchy stress tensor. We use the Kelvin-Voigt viscoelastic model with nonlinear elasticity, $\mathbf{T} = \mathbf{T}_v + \mathbf{T}_e$, where the viscous stress is Newtonian and the elastic stress includes a neo-Hookean term and pressure [37]. Four boundary conditions apply at the bubble surface [21]. Two are kinematic boundary conditions: no-penetration and no-slip,

$$\frac{\partial F}{\partial t} + (\mathbf{u}\cdot\nabla)F = 0, \tag{3}$$

$$\mathbf{u}_{t_1} = \mathbf{u}_{t_2}, \tag{4}$$

where the subscript t denotes the tangential component and the subscripts 1 and 2 represent the gas and medium phases, respectively. The other two are dynamic boundary conditions for normal and shear stresses,

$$\mathbf{n}\cdot\mathbf{T}_2\mathbf{n} - \mathbf{n}\cdot\mathbf{T}_1\mathbf{n} = S\nabla\cdot\mathbf{n}, \tag{5}$$

$$\mathbf{n}\times\mathbf{T}_2\mathbf{n} - \mathbf{n}\times\mathbf{T}_1\mathbf{n} = 0, \tag{6}$$

where $\mathbf{n} = \nabla F/|\nabla F|$ is the normal unit vector at the bubble surface and S is the surface tension. Note that we solve Cauchy's equation of motion for the surrounding medium for which the bubble pressure is incorporated through the dynamic boundary condition.

2.2. Field decomposition

For simplicity, we assume that the Mach numbers corresponding to the bubble oscillations are sufficiently low that the flow is incompressible in the surroundings; this assumption is reasonable in the near field where the strains are large and could readily be relaxed by accounting for compressibility with an effective viscosity [40]. Since velocity changes due to non-spherical perturbations are much smaller than those due to radial motion, we decompose the velocity field as follows [21],

$$\mathbf{u} = \mathbf{u}_0 + \epsilon\,\mathbf{u}_p + \epsilon\,\mathbf{u}_r, \tag{7}$$

where the subscripts 0 and p denote radial motion and non-spherical corrections, respectively, which are both irrotational, r denotes the rotational correction, and ϵ is of order $O(|a_n|/R)$. We refer to the last term not as a viscous correction but as a rotational correction, because even if the flow is incompressible and irrotational there is a viscous contribution from the bubble surface due to the jump condition of normal stress (5). The pressure field is decomposed accordingly,

$$p = p_0 + \epsilon\,p_p + \epsilon\,p_r. \tag{8}$$

After the field decomposition, Cauchy's equation of motion (2) can be split into two parts, the first for the irrotational component:

$$\frac{\partial}{\partial t}(\mathbf{u}_0 + \epsilon\,\mathbf{u}_p) + [(\mathbf{u}_0 + \epsilon\,\mathbf{u}_p)\cdot\nabla](\mathbf{u}_0 + \epsilon\,\mathbf{u}_p) = \frac{1}{\rho}\mathrm{div}(\mathbf{T}_{\mathbf{v_{irrot}}} + \mathbf{T}_e), \tag{9}$$

where $\mathbf{T}_{\mathbf{v_{irrot}}}$ is the stress contribution to the irrotational velocity components $\mathbf{u}_0 + \epsilon\,\mathbf{u}_p$, and the pressure $p_0 + \epsilon\,p_p$ is included in $\mathbf{T}_e$ as the pseudo-pressure $\mathscr{P}$ [37]. Since the flow is incompressible and

irrotational, the divergence of $\mathbf{T}_{\mathbf{v}_{\mathbf{irrot}}}$ vanishes, and the velocity potential ϕ, upon satisfying the no-penetration condition (3), is given by [20],

$$\phi = \frac{R^2\dot{R}}{r} + \frac{R^{n+2}}{(n+1)r^{n+1}}\left(\dot{a}_n + \frac{2\dot{R}}{R}a_n\right)Y_{nm}. \tag{10}$$

Then, the velocity is obtained from $\mathbf{u}_0 + \epsilon\,\mathbf{u}_p = -\nabla\phi$. The pseudo-pressure is determined by integrating Eq. (9) from the bubble surface to infinity [33],

$$\mathscr{P} = p_\infty + \rho\left[R\ddot{R} + \tfrac{3}{2}\dot{R}^2 + \tfrac{3\dot{R}\dot{a}_n + R\ddot{a}_n - (n-1)\dot{R}a_n}{n+1}Y_{nm}\right] + \tfrac{G}{2}\left[5 - 4\left(\tfrac{R_0}{R}\right) + \left(\tfrac{R_0}{R}\right)^4\right] + \tfrac{G}{R}\left[2\tfrac{R_0}{R} - 2\left(\tfrac{R_0}{R}\right)^4 + \tfrac{n(n+1)}{1+\frac{R_0}{R}+\left(\frac{R_0}{R}\right)^2}\right]a_nY_{nm}, \tag{11}$$

where $p_\infty(t)$ is the time-dependent far-field pressure and R_0 is the equilibrium bubble radius in a strain-free state. The far-field pressure consists of atmospheric pressure and acoustic pressure $p_\infty(t) = p_{atm} + p_a(t)$,
where the acoustic pressure can be assumed uniform in space provided that the wavelength is sufficiently large compared to the bubble size.

The second part of Cauchy's equation of motion is for the rotational component, which can be obtained by subtracting Eq. (9) from Eq. (2), thus yielding,

$$\frac{\partial}{\partial t}(\epsilon\,\mathbf{u}_r) + (\mathbf{u}_0\cdot\nabla)\epsilon\,\mathbf{u}_r + (\epsilon\,\mathbf{u}_r\cdot\nabla)\mathbf{u}_0 = \frac{1}{\rho}\mathrm{div}\mathbf{T}_{\mathbf{v}_{\mathbf{rot}}}, \tag{12}$$

where $\mathbf{T}_{\mathbf{v}_{\mathbf{rot}}}$ is the stress contribution to the rotational velocity component $\epsilon\,\mathbf{u}_r$ and the pressure $\epsilon\,p_r$. For a purely viscous surrounding medium, Prosperetti [21] solved Eq. (12) and determined the velocity and pressure field, which was also reviewed by Liu et al. [26],

$$\epsilon\,\mathbf{u}_r = \mathscr{T}Y_{nm}\mathbf{e}_r - \nabla\Phi, \tag{13}$$

$$\epsilon\,p_r = \mu\frac{n}{R}\mathscr{T}(R,t)Y_{nm} + \rho n\frac{\dot{R}}{R}Y_{nm}\int_R^\infty\left[\left(\frac{R}{s}\right)^3 - 1\right]\left(\frac{R}{s}\right)^n\mathscr{T}(s,t)ds, \tag{14}$$

where

$$\Phi = Y_{nm}\left[\left(\alpha(t) + \tfrac{n+1}{2n+1}\int_R^r s^{-n}\mathscr{T}(s,t)ds\right)r^n + \left(\tfrac{n}{n+1}R^{2n+1}\alpha(t) + \tfrac{n}{2n+1}\int_R^r s^{n+1}\mathscr{T}(s,t)ds\right)r^{-(n+1)}\right], \tag{15}$$

$$\alpha(t) = -\frac{n+1}{2n+1}\int_R^\infty s^{-n}\mathscr{T}(s,t)ds. \tag{16}$$

The toroidal component of the vorticity field $\mathscr{T}(r,t)$ is determined by the second-order partial differential equation (PDE),

$$\rho\frac{\partial\mathscr{T}}{\partial t} + \rho\frac{\partial}{\partial r}\left[\dot{R}\left(\frac{R}{r}\right)^2\mathscr{T}\right] - \mu\frac{\partial^2\mathscr{T}}{\partial r^2} + \mu n(n+1)\frac{\mathscr{T}}{r^2} = 0, \tag{17}$$

which requires two boundary conditions: $\mathscr{T} \to 0$ as $r \to \infty$, and shear stress continuity (6) at the bubble surface. Since the viscosity of the gas is much smaller than that of the surrounding medium, the shear stress continuity condition (6) reduces to a balance between viscous shear stress and elastic shear stress of the surrounding medium,

$$\mathscr{T}(R,t) = \frac{2(n+2)}{n+1}\dot{a}_n - \frac{2(n-1)}{n+1}\frac{\dot{R}}{R}a_n + \frac{G}{\mu}\left(\frac{R_0}{R}\right)^4 a_n - 2R^{n-1}\int_R^\infty\frac{\mathscr{T}(s,t)}{s^n}ds. \tag{18}$$

The toroidal component of the vorticity evolves according to Eq. (17), while the poloidal component is zero if its initial condition is zero [21].

2.3. The final form of the equations

The jump condition of normal stress (5) reduces to

$$-\epsilon\,p_r + 2\mu\frac{\partial(\epsilon\,u_{r,r})}{\partial r} + 2\mu\frac{\partial(u_{0,r} + \epsilon\,u_{p,r})}{\partial r} + G\left(\frac{r_0}{r}\right)^4 - \mathscr{P} + p_b = \frac{2S}{R} + \frac{(n-1)(n+2)}{R^2}Sa_nY_{nm}, \tag{19}$$

where $u_{0,r}$, $u_{p,r}$ and $u_{r,r}$ are the radial component of the corresponding velocity vectors and r_0 is the original configuration. The first two terms on the left-hand side are contributions from the rotational correction, while the third term is the viscous contribution, which still remains even assuming irrotational flow in the surrounding medium. Substituting the velocity and pressure $u_{0,r}$, $\epsilon\,u_{p,r}$, $\epsilon\,u_{r,r}$, $\mathscr{P}$ and $\epsilon\,p_r$ into Eq. (19) to order $O(1)$ gives the nonlinear ODE for mean bubble radius,

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{p_b - p_\infty}{\rho} - \frac{2S}{\rho R} - \frac{4\mu\dot{R}}{\rho R} - \frac{G}{2\rho}\left[5 - 4\frac{R_0}{R} - \left(\frac{R_0}{R}\right)^4\right], \tag{20}$$

which is the Rayleigh-Plesset equation accounting for the nonlinear elasticity of the surrounding medium [37]. To order $O(a_n)$, the linear integro-differential equation for the non-spherical mode n of amplitude

a_n is

$$\ddot{a}_n + (A_n + A_{n_{rot}})\dot{a}_n + (B_n + B_{n_{rot}})a_n + C_{n_{rot}} = 0, \tag{21}$$

where

$$A_n = 3\frac{\dot{R}}{R} + 2(n+1)(n+2)\frac{\mu}{\rho R^2}, \tag{22}$$

$$B_n = -(n-1)\frac{\ddot{R}}{R} + 4(n-1)(n+1)\frac{\mu\dot{R}}{\rho R^3} + (n-1)(n+1)(n+2)\frac{S}{\rho R^3} + (n+1)\frac{G}{\rho R^2}\left[2\frac{R_0}{R} + 2\left(\frac{R_0}{R}\right)^4 + \frac{n(n+1)}{1+\frac{R_0}{R}+\left(\frac{R_0}{R}\right)^2}\right], \tag{23}$$

$$A_{n_{rot}} = -2n(n+1)(n+2)\frac{\mu}{\rho R^2}, \tag{24}$$

$$B_{n_{rot}} = 2n(n-1)(n+1)\frac{\mu\dot{R}}{\rho R^3} - n(n+1)^2\frac{G}{\rho R^2}\left(\frac{R_0}{R}\right)^4, \tag{25}$$

$$C_{n_{rot}} = n(n+1)(n+2)\frac{\mu}{\rho R^2}\mathscr{T}(R,t) - n(n+1)\frac{\dot{R}}{R^2}\int_R^\infty\left[1-\left(\frac{R}{s}\right)^3\right]\left(\frac{R}{s}\right)^n\mathscr{T}(s,t)ds. \tag{26}$$

The coefficients in the harmonic oscillator of Eq. (21) are expressed as the sum of an irrotational term and a rotational correction. Compared to the model of Prosperetti [21], the novelty in this work lies in the elastic terms in Eqs. (18) (third term), (23) (last term), and (25) (last term). Since terms of order $O(a_n^2)$ and higher are neglected, the equations for the different modes are not coupled and do not influence the base flow R. Although the equations are linearized in a_n, the oscillations of the base flow R may be nonlinear. Furthermore, the model includes nonlinear elastic effects. Shape stability of a bubble surrounded by linear and nonlinear elastic media is investigated in a separate work [33], which shows that nonlinear elasticity gives rise to a more stable spherical shape than linear elasticity. When $G = 0$, our model reduces to the purely viscous model of Prosperetti [21].

In summary, the non-spherical bubble surface in a viscoelastic medium is determined by solving two ODEs: one for the mean bubble radius (20), and the other for the non-spherical mode amplitudes (21). The bubble pressure is assumed homobaric and determined by another ODE (Appendix A.1), where the temperature inside the bubble is calculated by solving a PDE for energy balance (Appendix A.2); alternately, the process could be assumed polytropic, though it would be less accurate in the case of nonlinear oscillations [41]. The toroidal field is used to determine $C_{n_{rot}}$ by solving the PDE (17). Since Eq. (21) does not explicitly depend on the driving pressure, non-spherical mode instability is induced by the unsteady mean flow.

The model is solved numerically as follows. We use the Cash-Karp Runge-Kutta method [42] for the time integration of the two ODEs (20) and (21). For the PDE (17), we apply the same numerical treatment as Liu et al. [26]: Eq. (17) is transformed into a fixed computational domain by the change of variables, $y = r/R(t)$, the convection and diffusion terms are discretized by upwind and central differences, and the standard fourth-order Runge-Kutta method is used for the time integration. We use Simpson's rule for the integral terms in Eqs. (18) and (26). The computational domain is set to $1.00 \leqslant y \leqslant y_{\max} = 2.00$ with uniform grid size $\Delta y = 0.01$, which is large enough to capture the vorticity field (see Section 3). We use a sufficiently small time step to satisfy the Courant-Friedrichs-Lewy (CFL) condition. The dimensionless form of our model is summarized in Appendix B.

3. Damping rate

We first investigate the role of viscosity on non-spherical bubble dynamics, which will help us understand the damping and amplitude threshold of the parametric instability in Section 4. We further explain the contributions of rotational and irrotational fields to the damping rate in the surroundings and at the bubble surface. $A_{n_{rot}}$, $B_{n_{rot}}$ and $C_{n_{rot}}$ in Eq. (21) are rotational correction terms, which come from the first two terms on the left-hand side of Eq. (19). With no rotational correction, our model reduces to the irrotational viscoelastic model of Murakami et al. [43]. The deformation of the no-slip bubble surface, where shear stress must be continuous, generates vorticity diffusing into the surrounding medium. From a mathematical viewpoint, the rotational field requires the additional boundary condition (18) to solve the second-order PDE (17).

We conduct a test simulation to examine the damping rate. We consider a bubble with $R_0 = 100\ \mu$m in water of density $\rho = 998$ kg/m^3, surface tension $S = 0.072$ N/m, viscosity $\mu = 1.0$ mPa·s and no shear modulus ($G = 0$ kPa), under atmospheric pressure. We prescribe an ultrasound pressure wave with driving frequency $f_d = 28$ kHz and pressure amplitude $p_a = 3.5$ kPa. The initial conditions are set to $R = R_0$, $\dot{R} = 0$, $a_n = 0.01R_0$, $\dot{a}_n = 0$ and $\mathscr{T}(r, t = 0) = 0$. Fig. 2 shows the solution of our model including the time histories of far-field pressure, mean bubble radius and mode amplitudes ($n = 2$ to $n = 6$) normalized by the mean bubble radius. Mode $n = 4$ exhibits growth and parametric resonance, as explained in greater detail in Section 4. In Fig. 3, we compare mode $n = 4$ for different cases: the solution with the irrotational model, the solution with different computational domain sizes $y_{\max} = 1.00$ (the integral terms are zero), $y_{\max} = 2.00$ and $y_{\max} = 3.00$. Our model exhibits smaller growth than the irrotational model. This behavior follows from the local damping due to viscous dissipation. Substituting the shear stress continuity condition (18) into Eq. (26), the damping rate is obtained from Eq. (21),

$$\zeta_n = 2(n+1)(n+2)\frac{\mu}{\rho R^2} + \zeta_{n_{rot}}, \tag{27}$$

with rotational correction

$$\zeta_{n_{rot}} = 2n(n+2)\frac{\mu}{\rho R^2}, \tag{28}$$

which indicates that the rotational field increases the damping rate. The rotational correction term $\zeta_{n_{rot}}$ stems from satisfying the shear stress continuity at the bubble surface.

The solution converges with relatively small computational domain $y_{\max} = 2.00$. In this problem, the boundary layer tends to a finite size, which is thin compared to the mean bubble radius. This behavior is expected given that the interface oscillation gives rise to a problem analogous to Stokes' second problem. Fig. 3 also shows that the vorticity field (integral terms) has a destabilizing effect on the bubble shape. However, the converged solution still exhibits smaller growth than the irrotational model. In fact, the main contribution to the rotational field originates from shear stress continuity at the bubble surface rather than the rotational field itself in the surroundings. The shear stress continuity condition increases the damping rate. Furthermore, the elasticity in the shear stress continuity condition contributes to parametric instability.

4. Parametric instability

Parametric instability of a bubble driven by continuous forcing is described by the onset of non-spherical oscillations. This instability is caused by parametric resonance between non-spherical modes and the mean bubble radius under simple harmonic forcing, and has previously

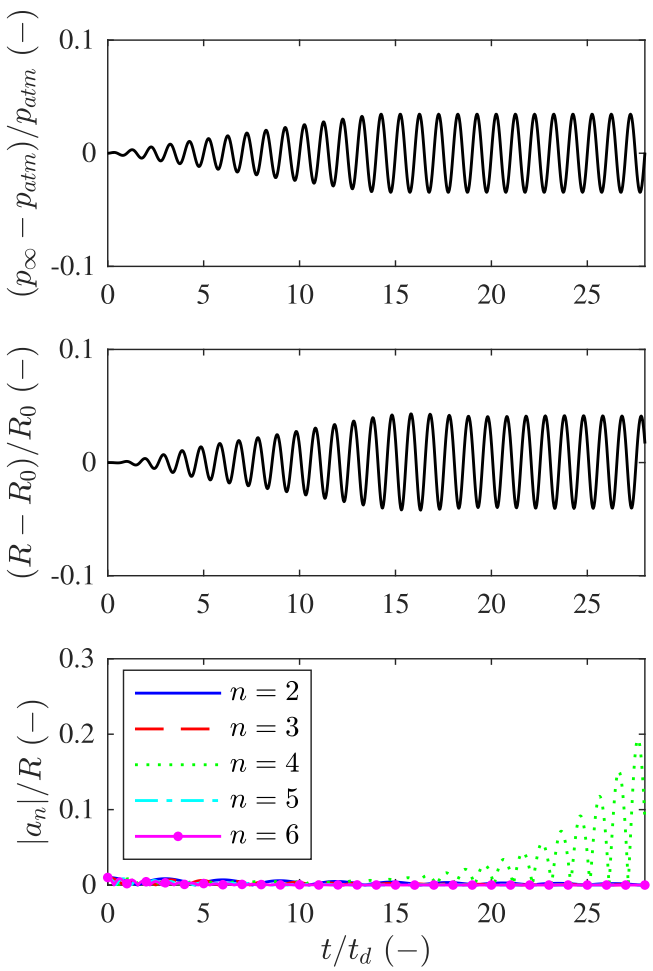

Fig. 2. Solution for ultrasound-induced bubble oscillations in water: the time histories of far-field pressure (top), mean bubble radius (middle) and non-spherical mode amplitudes (bottom). $t_d = 1/f_d$ is the characteristic time. Mode $n = 4$ is unstable because of parametric resonance.

been investigated for bubble oscillations in water [22,17]. Mathematically, the instability is described by a Mathieu equation [44,24,26], in which the non-spherical mode becomes unstable when $\omega_d = 2\omega_n$ where $\omega_d = 2\pi f_d$ is the driving frequency and ω_n is the natural frequency of each non-spherical mode. The theory describing parametric instabilities of bubbles in liquids was developed for Newtonian surrounding fluids. Since soft materials exhibit viscoelastic behavior, we extend the theory to account for viscoelasticity. In this section, we derive the natural frequency and predict the most unstable mode for a bubble in a viscoelastic medium. Furthermore, we derive the Mathieu equation corresponding to our model (21) and predict the parametric instability based on the amplitude threshold of radial motion.

4.1. The most unstable mode

We consider an initially spherical oscillating bubble. It is reasonable to assume $\mathscr{T}(r \neq R, t) = 0$ as long as no vorticity is present around the bubble before the non-spherical mode appears. Substituting (18) into (26) yields

$$C_{n_{rot}} \approx 2n(n+2)^2\frac{\mu}{\rho R^2}\dot{a}_n - 2n(n-1)(n+2)\frac{\mu\dot{R}}{\rho R^3}a_n + n(n+1)(n+2)\frac{G}{\rho R^2}\left(\frac{R_0}{R}\right)^4 a_n. \tag{29}$$

Then, our model (21) reduces to

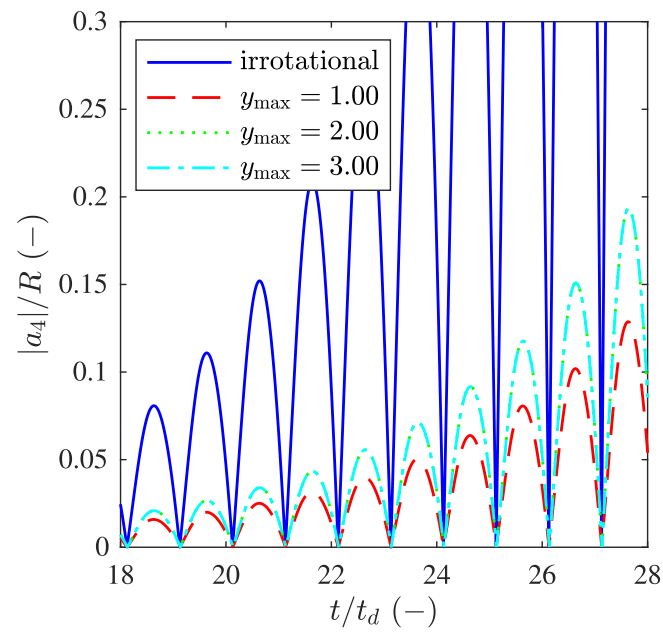

Fig. 3. Comparison of the time histories of mode $n = 4$ during the ultrasound-induced bubble oscillations for different cases: irrotational model, our model with different computational domain sizes $y_{\max} = 1.00$, $y_{\max} = 2.00$ and $y_{\max} = 3.00$. The solution for $y_{\max} = 3.00$ is indistinguishable from that for $y_{\max} = 2.00$. The full model taking into account the rotational field exhibits smaller growth than the irrotational model.

$$\ddot{a}_n + (A_n + A'_{n_{rot}})\dot{a}_n + (B_n + B'_{n_{rot}})a_n = 0, \tag{30}$$

where

$$A'_{n_{rot}} = 2n(n+2)\frac{\mu}{\rho R^2}, \tag{31}$$

$$B'_{n_{rot}} = -2n(n-1)\frac{\mu\dot{R}}{\rho R^3} + n(n+1)\frac{G}{\rho R^2}\left(\frac{R_0}{R}\right)^4. \tag{32}$$

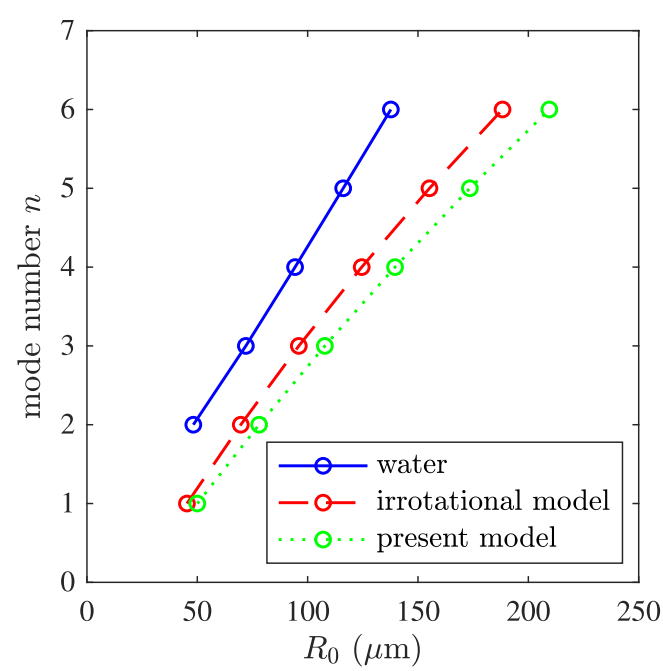

Fig. 4. Relation between the most unstable mode n and equilibrium bubble radius R_0 in parametric instability for a fixed driving frequency $f_d = 28$ kHz given by Eq. (33).

Since this is a second-order linear ODE, the square of its natural frequency corresponds to $B_n + B'_{n_{rot}}$. Ignoring the transient variation in R, the n-th order natural frequency is obtained as

$$\omega_n^2 = (n-1)(n+1)(n+2)\frac{S}{\rho R_0^3} + (n+1)\frac{G}{\rho R_0^2}\left[4 + \frac{n(n+1)}{3}\right] + \omega_{n_{rot}}^2, \tag{33}$$

with rotational correction

$$\omega_{n_{rot}}^2 = n(n+1)\frac{G}{\rho R_0^2}, \tag{34}$$

which indicates that the shear modulus of the surrounding medium increases the natural frequency of the non-spherical mode. This behavior is consistent with the notion that an elastic force has a restoring effect and is similar to the effect of an elastic membrane on an encapsulated bubble [26]. Eq. (33) also implies that the shear modulus of the surrounding medium enables parametric resonance of mode $n = 1$, which does not exist in water because translational oscillations cannot be induced by surface tension or an elastic membrane. With no rotational correction term $\omega_{n_{rot}}^2$, Eq. (33) reduces to the n-th order natural frequency for the irrotational viscoelastic model [43]. When $G = 0$, Eq. (33) reduces to the n-th order natural frequency for a bubble in water [45,21], which has been validated by experimental observations of parametric resonance [17]. Similar to the damping rate (27), the rotational correction term stems from satisfying shear stress continuity at the bubble surface.

For a fixed driving frequency, Eq. (33) relates the most unstable mode n to the equilibrium bubble radius R_0 as an indication for the parametric instability. As an example, Fig. 4 shows this relation for $f_d = 28$ kHz. Mode $n = 4$ is the most unstable for a bubble of $R_0 = 100\ \mu$m in water, as already observed in the test simulation (Fig. 2). Fig. 4 also includes the relation for a viscoelastic medium with surface tension $S = 0.040$ N/m and shear modulus $G = 1.7$ kPa, where the material properties are of gelatin gel from the experiments of Hamaguchi and Ando [18]. For a bubble in water, the most unstable mode depends only on surface tension. In a viscoelastic medium on the other hand, the most unstable mode varies depending on the shear modulus. For a given bubble size, shear modulus reduces the most unstable mode number to a value below that of water or that predicted by the irrotational model.

4.2. Amplitude threshold

We can find the amplitude threshold of radial motion for the parametric instability by reducing our model (21) to a Mathieu equation. As explained in Section 2.3, the instability is induced by the mean bubble oscillations. Francescutto and Nabergoj [44] reduced the purely viscous model to a Mathieu equation and found that the parametric instability occurs when the mean bubble radius exceeds a threshold value. Following a similar procedure, we derive the amplitude threshold corresponding to a bubble in a viscoelastic medium. We introduce the change of variables:

$$a_n = b_n \exp\left[-\frac{1}{2}\int (A_n + A'_{n_{rot}})dt\right], \tag{35}$$

such that Eq. (30) can be transformed to

$$\ddot{b}_n - (\mathscr{G}_n + \mathscr{G}_{n_{rot}})b_n = 0, \tag{36}$$

where

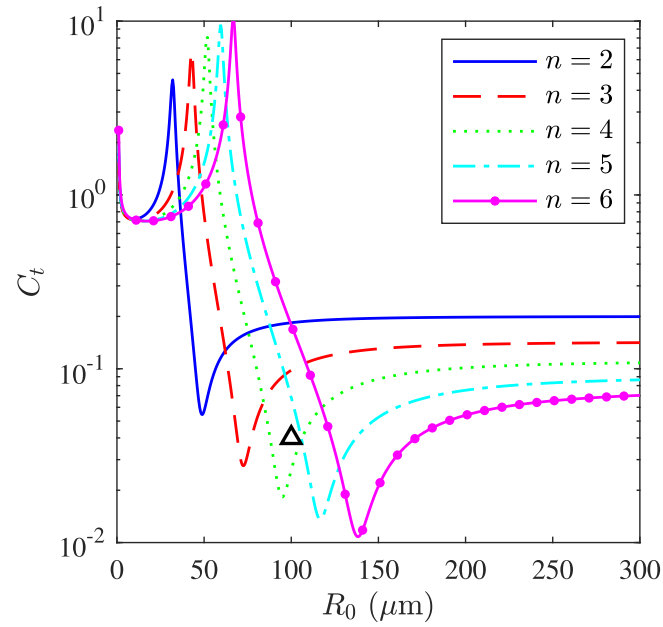

Fig. 5. Amplitude threshold C_t versus equilibrium bubble radius R_0 for water given by Eq. (51) with driving frequency $f_d = 28$ kHz. The triangle corresponds to the simulation from Section 3, which is inside the unstable region of mode $n = 4$.

$$\mathscr{G}_n = \frac{3}{4}\left(\frac{\dot{R}}{R}\right)^2 + \left(n + \frac{1}{2}\right)\frac{\ddot{R}}{R} - (n-1)(n+1)(n+2)\frac{S}{\rho R^3} - 6n(n+1)\frac{\mu\dot{R}}{\rho R^3} + \left[(n+1)(n+2)\frac{\mu}{\rho R^2}\right]^2 - (n+1)\frac{G}{\rho R^2}\left[2\frac{R_0}{R} + 2\left(\frac{R_0}{R}\right)^4 + \frac{n(n+1)}{1 + \frac{R_0}{R} + \left(\frac{R_0}{R}\right)^2}\right], \tag{37}$$

$$\mathscr{G}_{n_{rot}} = -6n\frac{\mu\dot{R}}{\rho R^3} + 2n(n+1)(n+2)^2\left(\frac{\mu}{\rho R^2}\right)^2 + \left[n(n+2)\frac{\mu}{\rho R^2}\right]^2 - n(n+1)\frac{G}{\rho R^2}\left(\frac{R_0}{R}\right)^4. \tag{38}$$

Considering that the radial motion driven by an ultrasound wave is simple harmonic,

$$R = R_0(1 + C\cos\omega_d t), \tag{39}$$

where the relative amplitude $C = |R - R_0|/R_0$ is of order $O(|a_n|/R)$. Substituting Eq. (39) into Eq. (36) and using the change of variables, $z = \omega_d t/2$, a Mathieu equation is obtained to order $O(a_n)$

$$\frac{d^2 b_n}{dz^2} + [\beta_0 + 2\beta_2\cos(2z - \varphi)]b_n = 0, \tag{40}$$

where

$$\beta_0 = \alpha_1 - \left(\alpha_2 + \alpha_{2_{rot}}\right)^2 + \alpha_3\left[4 + \frac{n(n+1)}{3}\right] + \alpha_{3_{rot}}, \tag{41}$$

$$\beta_2 = C\sqrt{X^2 + Y^2}, \tag{42}$$

$$\varphi = \arctan\frac{Y}{X}, \tag{43}$$

and

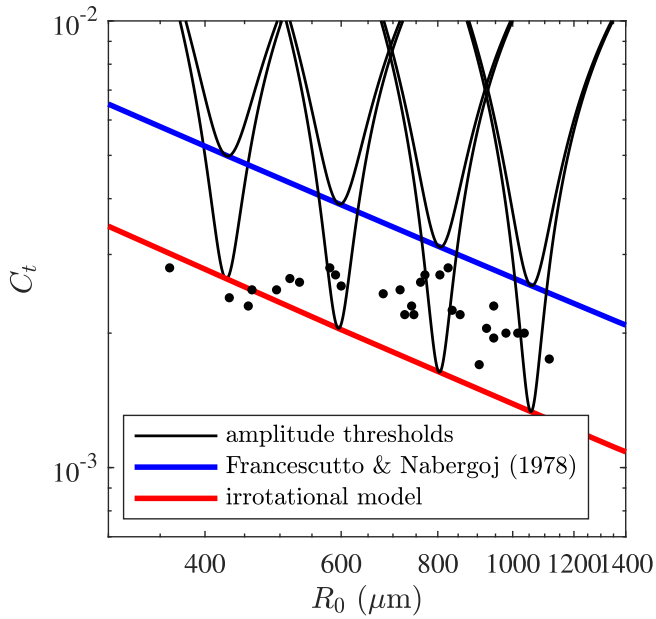

Fig. 6. Amplitude thresholds C_t versus equilibrium bubble radius R_0 for water given by Eq. (51), where the driving frequency is set to the zeroth-order natural frequency (52). The minima of the amplitude thresholds correspond to mode $n = 8$ to $n = 11$ from left to right. The experiments of Hullin [47] are shown by black circles, which are mostly bounded by lines following Francescutto and Nabergoj [44] (blue line) and the irrotational model (red line).

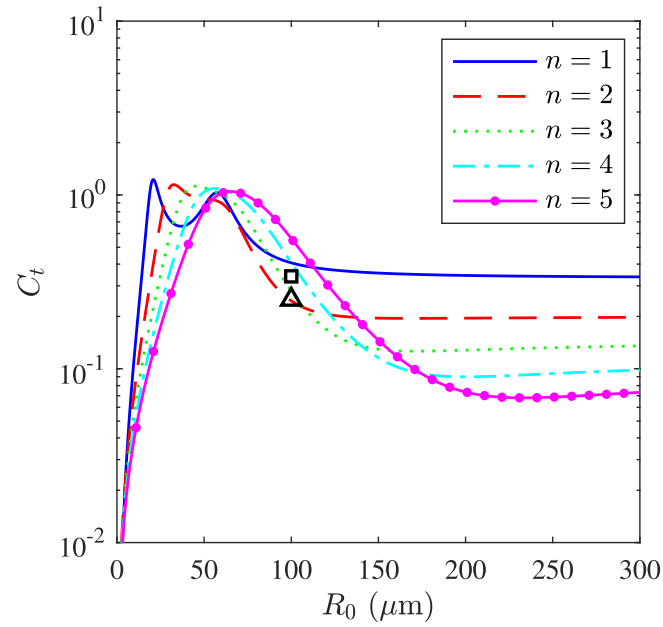

Fig. 7. Amplitude threshold C_t versus equilibrium bubble radius R_0 for a viscoelastic medium given by Eq. (51) with driving frequency $f_d = 28$ kHz. The triangle and square correspond to the experiment of Hamaguchi and Ando [18] and the example in Section 5.2, respectively, which are in the vicinity of the unstable regions of modes $n = 2$ and $n = 3$.

$$X = (2n+1) - \frac{3}{2}\alpha_1 + 2\left(\alpha_2 + \alpha_{2_{rot}}\right)^2 - \alpha_3\left[18 + \frac{n(n+1)}{3}\right] - 6\alpha_{3_{rot}}, \tag{44}$$

$$Y = \alpha_4 + \alpha_{4_{rot}}, \tag{45}$$

$$\alpha_1 = 4(n-1)(n+1)(n+2)\frac{S}{\rho\omega_d^2 R_0^3}, \tag{46}$$

$$\alpha_2 = 2(n+1)(n+2)\frac{\mu}{\rho\omega_d R_0^2}, \qquad \alpha_{2_{rot}} = 2n(n+2)\frac{\mu}{\rho\omega_d R_0^2}, \tag{47}$$

$$\alpha_3 = 4(n+1)\frac{G}{\rho\omega_d^2 R_0^2}, \qquad \alpha_{3_{rot}} = 4n(n+1)\frac{G}{\rho\omega_d^2 R_0^2}, \tag{48}$$

$$\alpha_4 = -12n(n+1)\frac{\mu}{\rho\omega_d R_0^2}, \qquad \alpha_{4_{rot}} = -12n\frac{\mu}{\rho\omega_d R_0^2}. \tag{49}$$

We again distinguish rotational corrections from irrotational terms. According to stability theory for the Mathieu equation (or the Hill equation more generally), the stability condition for the first unstable region is [46]

$$(\beta_0 - 1)^2 + 2(\beta_0 + 1)\delta^2 + \delta^4 > \beta_2^2, \tag{50}$$

where $2\delta = 2(\alpha_2 + \alpha_{2_{rot}})$ is the constant term in the series of $A_n + A'_{n_{rot}}$ in Eq. (30). The stability condition (50) results in

$$C < C_t = \sqrt{\frac{\left\{(\alpha_1 - 1) + \alpha_3\left[4 + \frac{n(n+1)}{3}\right] + \alpha_{3_{rot}}\right\}^2 + 4\left(\alpha_2 + \alpha_{2_{rot}}\right)^2}{X^2 + Y^2}}, \tag{51}$$

where C_t is the amplitude threshold of radial motion. In short, parametric instability occurs when the relative amplitude exceeds the amplitude threshold: $C \geqslant C_t$.

For a given driving frequency ω_d, mode number n, and equilibrium bubble radius R_0, the amplitude threshold C_t is determined by Eq. (51). As an example, Fig. 5 shows the amplitude threshold for modes $n = 2$ to $n = 6$ in water with a fixed driving frequency $f_d = 28$ kHz, where the same physical properties as in Section 3 are used. Each of these modes becomes unstable when the relative amplitude exceeds the corresponding amplitude threshold. The relative amplitude in the test simulation is $C \approx 0.04$ (Fig. 2), which is shown by a triangle in Fig. 5. This triangle is inside the unstable region for mode $n = 4$ but outside the unstable regions of the other modes, thus confirming that mode $n = 4$ is the only unstable mode under these conditions. Furthermore, the minima of the amplitude threshold in Fig. 5 and the most unstable mode in Fig. 4 correspond to the same equilibrium bubble radius. Higher modes $n > 6$ are stable, as their minima are located at larger equilibrium bubble radii.

For water, Francescutto and Nabergoj [44] compared the amplitude threshold for purely viscous surroundings with the experiments of Hullin [47], where the driving frequency was set to the zeroth-order natural frequency [48]:

$$\omega_0^2 = \frac{3\kappa}{\rho R_0^2}\left(p_\infty + \frac{2S}{R_0}\right) - \frac{2S}{\rho R_0^3}. \tag{52}$$

The polytropic index is set to $\kappa = 1.4$, which is valid for large bubbles [40]. In Fig. 6, we reproduce the amplitude threshold for water and compare against results with the irrotational model (Eq. (51) with no rotational correction). Since the second or third unstable regions provide additional minima [46], it is reasonable to define the amplitude threshold for all unstable regions as the line obtained by connecting the minima of the first unstable region as drawn in Fig. 6 [44]. The experimental data [47] are for the most part between lines following Francescutto and Nabergoj [44] (higher bound) and the irrotational model (lower bound). This behavior is expected from Fig. 3 in which the solution of the full model lies between the solution obtained with the full model without vorticity on a domain with $y_{max} = 1.00$ (smallest growth) and the solution obtained with the irrotational model (largest growth). Therefore, the precise threshold exists somewhere between Eq. (51) and the irrotational model depending on how the vorticity behaves. At small radii, a few experimental data points fall outside the predicted bounds. We believe this behavior may be due to contaminants

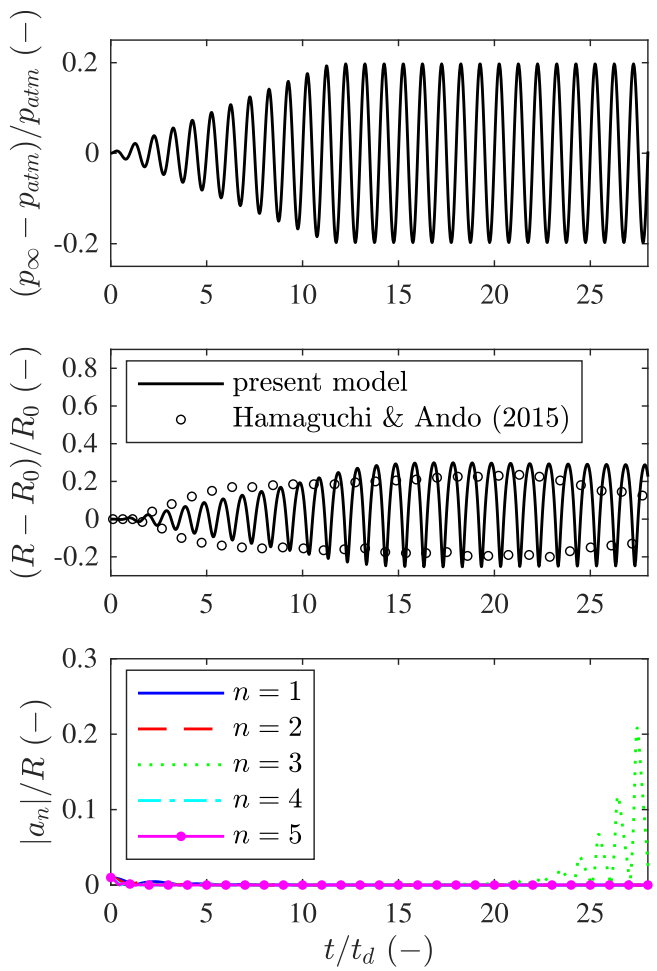

Fig. 8. Solution of Eqs. (20) and (21) compared to the experiments of Hamaguchi and Ando [18]: the time histories of far-field pressure (top), mean bubble radius (middle) and non-spherical mode amplitudes (bottom). Mode $n = 3$ becomes large after several oscillations due to parametric resonance.

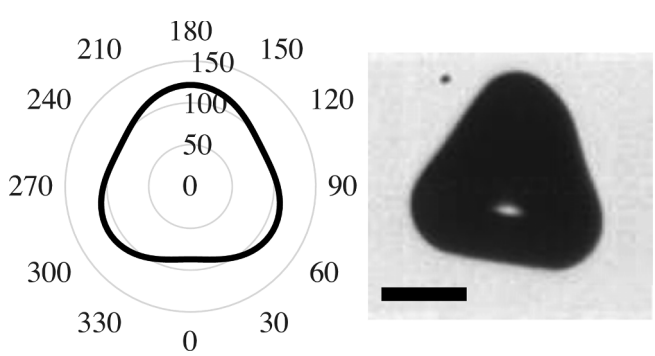

Fig. 9. Example of non-spherical bubble shape showing the unstable mode $n = 3$ observed in our solutions at $t/t_d = 27.5$ (left) and in the experiments at $t/t_d = 22.9$ (Reproduced from Hamaguchi and Ando [18], with the permission of AIP Publishing.).

in the water, which may increase the surface tension. Given their smaller surface area, smaller bubbles would be more sensitive to the presence of such contaminants. Verifying this conjecture would be a worthwhile investigation in the future.

Fig. 7 shows the amplitude threshold for modes $n = 1$ to $n = 5$ in a viscoelastic medium. We use a fixed driving frequency $f_d = 28$ kHz and the physical properties of gelatin gel with surface tension $S = 0.040$ N/m, viscosity $\mu = 14.1$ mPa·s and shear modulus

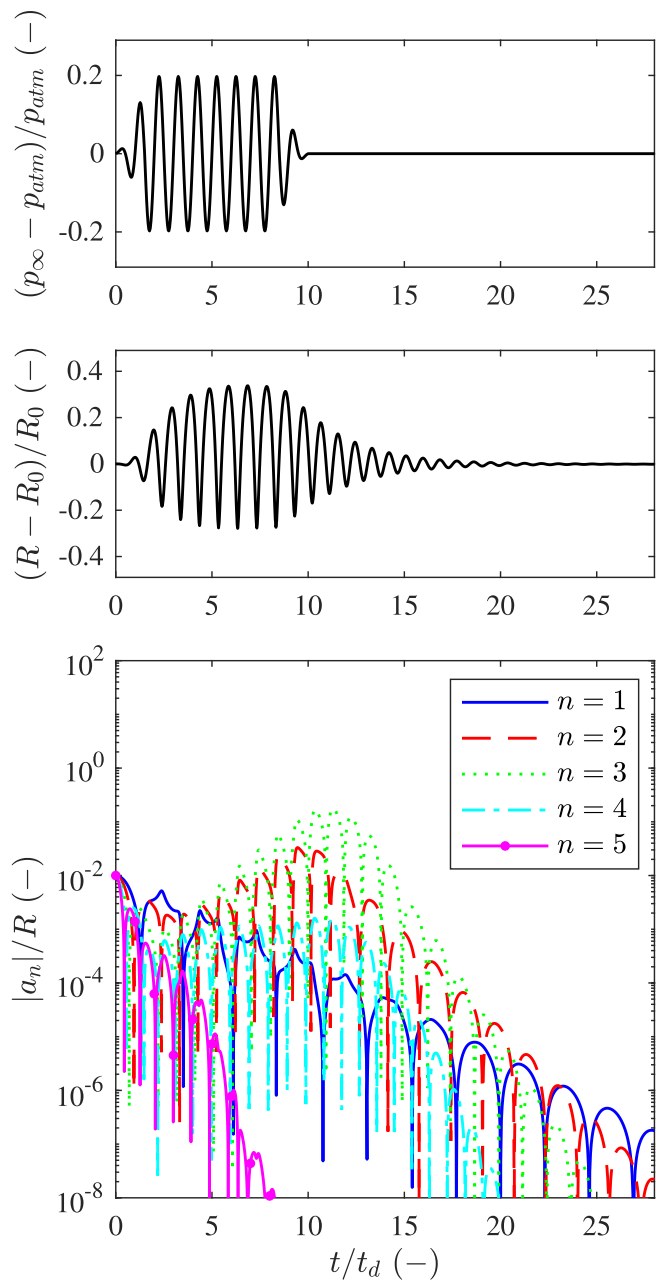

Fig. 10. Solution of Eqs. (20) and (21) under pulse-wave ultrasound irradiation: the time histories of far-field pressure (top), mean bubble radius (middle) and non-spherical mode amplitudes (bottom). Modes $n = 2$, $n = 3$ and $n = 4$ exponentially grow due to parametric resonance. After the passage of the pulse, higher modes decay faster than lower modes.

$G = 1.7$ kPa; these viscoelastic properties are inferred from experiments [18]. Compared to Fig. 5, viscoelasticity drastically changes the unstable region of the parametric instability. Viscosity smoothens the minima, while shear modulus shifts the minima toward larger equilibrium bubble radii. Hamaguchi and Ando [18] observed that mode $n = 3$ is the most unstable mode for a bubble with $R_0 = 100$ μm in the gelatin gel, which agrees with our prediction based on Fig. 4. The relative amplitude in their experiments is $C \approx 0.25$, which is shown by a triangle in Fig. 7. This point is close to the unstable regions of modes $n = 2$ and $n = 3$, while it is clearly outside the unstable regions of other modes. Although vorticity may slightly change these amplitude thresholds, Fig. 7 also implies parametric instability when mode $n = 3$ is unstable.

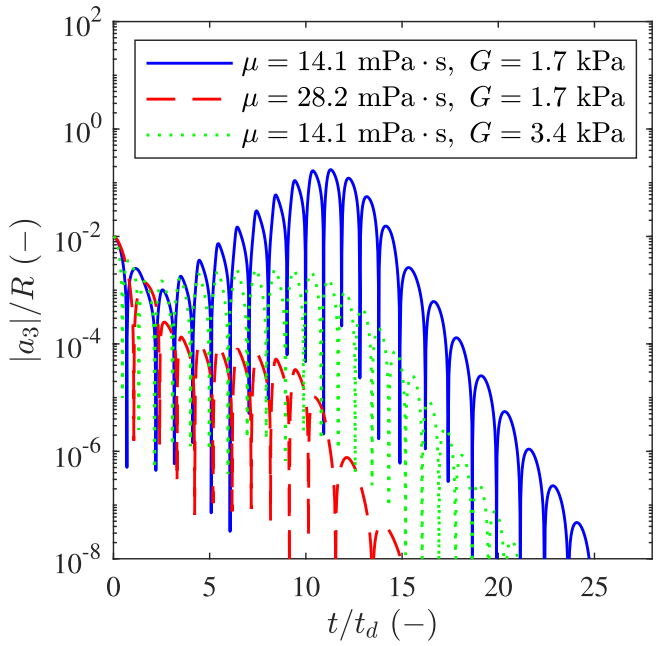

Fig. 11. Time history of the mode $n = 3$ in three different cases: the same properties as in Fig. 10, the same shear modulus but double viscosity, and the same viscosity but double shear modulus. Viscosity reduces the growth, while shear modulus changes the most unstable mode number to $n = 2$ for $G = 3.4$ kPa, thus making mode $n = 3$ less unstable.

5. Discussion

5.1. Validation

For validation purposes, we compare our results to the experiments of Hamaguchi and Ando [18]. Following their set-up, we consider a bubble with $R_0 = 100\ \mu$m in a 3 wt% gelatin gel with surface tension $S = 0.040$ N/m, viscosity $\mu = 14.1$ mPa·s and shear modulus $G = 1.7$ kPa, under atmospheric pressure. In the experiments, the laser-generated bubble creates a region of pre-strain around the bubble. However, the pre-strain is expected to be negligible in the case of 3 wt% gel [49]. We prescribe an ultrasound pressure waveform with driving frequency $f_d = 28$ kHz and pressure amplitude $p_a = 20$ kPa. The initial conditions are set to $R = R_0$, $\dot{R} = 0$, $a_n = 0.01R_0$, $\dot{a}_n = 0$ and $\mathscr{T}(r, t = 0) = 0$. The main uncertainty is the initial mode amplitudes, which are too small to resolve in the experiments. Versluis et al. [17] measured this initial condition to be $a_4 \approx 0.1\ \mu$m for $R_0 = 33\ \mu$m in water. Our initial perturbation is slightly larger than that of Versluis *et al.* [17] because surface tension effects on the initial perturbation are likely less important for the larger bubbles under consideration here, though of course this value is likely to vary from one experiment to another.

Fig. 8 shows the time histories of far-field pressure, mean bubble radius and mode amplitudes ($n = 1$ to $n = 5$) normalized by the mean bubble radius. The amplitude of the non-spherical modes is small, such that the oscillations are effectively spherical. However, after several oscillations, mode $n = 3$ has grown sufficiently due to parametric resonance that causes the bubble shape to visibly depart from spherical, while other non-spherical modes remain small. This behavior is predicted by our model by considering the natural frequency (Fig. 4) and amplitude thresholds (Fig. 7). Fig. 9 shows an example of the non-spherical bubble shape observed in our simulations at $t/t_d = 27.5$ and in the experiments [18] at $t/t_d = 22.9$, where both yield an unstable mode $n = 3$. These bubbles are two-dimensional projections from the perpendicular to the zenith direction ($m = 0$), where the axis of symmetry coincides with the initial plasma elongation direction in experiments. The discrepancy of the time at which mode $n = 3$ appears originates from the uncertainty in the initial mode amplitudes. It is also affected by the imperfect agreement in the mean bubble radius, which may come from uncertainties pertaining to the ultrasound pressure waveform [18]. In the experiments, the mean bubble oscillations are damped after the non-spherical mode manifest because the kinetic energy of the radial mode is transferred to the non-spherical mode and dissipated [50]. This kind of mode interaction is not captured in our linearized model. We also note that the model is valid provided that the mode amplitudes remain much smaller than the mean bubble radius ($|a_n| \ll R$). The model predicts the onset of the shape instability; it is not meant to describe the bubble dynamics after the non-spherical oscillations become significant.

5.2. The role of viscoelasticity on parametric instability

In medical applications, ultrasound waveforms generally have a finite pulse width [12]. In this section, we investigate the shape stability of a bubble under pulse-wave ultrasound irradiation and examine the role of viscoelastic properties (viscosity, shear modulus) on the time evolution of non-spherical mode amplitudes, with the same parameters as in Section 5.1. Fig. 10 shows the time histories of the far-field pressure, mean bubble radius and mode amplitudes ($n = 1$ to $n = 5$). Modes $n = 2$ and $n = 3$ grow, mode $n = 4$ slightly grows, while modes $n = 1$ and $n = 5$ decay exponentially. This behavior is consistent with the fact that the most unstable mode number is between 2 and 3 ($n = 2.7$ in Fig. 4). This behavior is further supported by the amplitude threshold, which is $C \approx 0.34$ and depicted by a square in Fig. 7, which lies inside the unstable regions of modes $n = 2$ and $n = 3$, close to the unstable region of mode $n = 4$, but outside the unstable regions of the other modes. After the passage of the pulse, every mode decays, with the higher modes decaying faster than the lower modes, as expected from the increasing damping rate (27) with increasing mode number.

We examine the effect of viscosity and shear modulus on non-spherical mode amplitudes by comparing the mode $n = 3$ presented in Fig. 10 with solutions where the viscosity is doubled (but shear modulus kept the same) and another where the shear modulus is doubled (but the viscosity kept the same), as shown in Fig. 11. The mean bubble radius is slightly different in each case, but this discrepancy has only a small influence on the mode amplitudes in this example. Viscosity increases the damping rate (27), thus reducing the growth of the mode and increasing the decay rate. Shear modulus increases the n-th order natural frequency (33), where the most unstable mode number becomes $n = 2$ for $G = 3.4$ kPa, thus making mode $n = 3$ less unstable. This analysis could be used to measure viscoelastic properties of soft materials at high rates and possibly large strains by experimentally observing the decay rate and unstable mode in shape oscillations of the bubble under pulse-wave ultrasound irradiation.

6. Conclusions

We developed a theory to determine the shape stability of a bubble in a soft solid described by a Kelvin-Voigt viscoelastic constitutive relation. The viscoelastic properties of the surrounding medium significantly influence the shape stability: viscosity increases the damping rate, thus suppressing the shape instability, while shear modulus increases the natural frequency, which changes the unstable mode in the parametric instability. The main contribution to the rotational field originates from shear stress continuity. The damping rate and unstable mode change depending on the viscoelastic properties on the stress conditions.

Given the driving waveform and the material properties of the surrounding medium, our model predicts the onset of the shape instability, as well as the most unstable mode, in agreement with published experimental results of the shape oscillations of a bubble in a

gelatin gel. We also derived the Mathieu equation and amplitude threshold for radial motion, which yields a stability diagram for any given driving frequency, surface tension and viscoelastic properties. Not only does our model predict shape instability, but it could be used to measure viscoelastic properties of soft materials based on the experimentally observed shape oscillations of a bubble.

In the future, additional experimental studies using different viscoelastic parameters would be helpful to bound uncertainties in our model. Although our study focused on the parametric instability induced by an ultrasound wave, it could be applied to instability during inertial bubble collapse [22], an important phenomenon in sonochemistry and biomedical applications [51,39]. In addition, our approach could be extended to study two interacting bubbles or bubble dynamics near rigid wall [52,27] in a viscoelastic medium where a jet is induced, and could be extended to account for mode interactions [53].

CRediT authorship contribution statement

Kazuya Murakami: Methodology, Software, Investigation, Writing - original draft. **Renaud Gaudron:** Methodology. **Eric Johnsen:** Conceptualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The model for bubble pressure

The gas inside the bubble (air) behaves in a homobaric fashion according to the following ODE [41]:

$$\dot{p}_b = \frac{3}{R}\left[-\gamma p_b \dot{R} + (\gamma - 1)K\frac{\partial T}{\partial r}\bigg|_{r=R}\right], \tag{A.1}$$

where γ is the specific heats ratio, $K(T) = AT + B$ is the thermal conductivity of the gas, and A and B are constants [41]. The temperature field $T(r, t)$ inside the bubble is determined by solving the PDE for energy balance:

$$\rho_g C_p\left(\frac{\partial T}{\partial t} + u_g\frac{\partial T}{\partial r}\right) = \dot{p}_b + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 K\frac{\partial T}{\partial r}\right), \tag{A.2}$$

where ρ_g is the gas density following the ideal gas law $p_b = \rho_g \mathscr{R}_g T$ where $\mathscr{R}_g$ is the gas constant, and C_p is the specific heat at constant pressure. The velocity field inside the bubble $u_g(r, t)$ is given by

$$u_g = \frac{1}{\gamma p_b}\left[-\frac{1}{3}r\dot{p}_b + (\gamma - 1)K\frac{\partial T}{\partial r}\right]. \tag{A.3}$$

The boundary conditions are $\partial T/\partial r|_{r=0} = 0$ at the origin and the cold liquid assumption $T(R, t) = T_\infty$ at the bubble wall [41]. The model is solved numerically: we semi-discretize the PDE by finite differences and use the Cash-Karp Runge–Kutta method [42] for time integration. The physical properties are summarized in Table 1.

Table 1
Values of the physical properties used for bubble pressure calculation.

Property	Value	Property	Value
γ	1.4	C_p	1.00 kJ/kg·K
$\mathscr{R}_g$	0.287 kJ/kg·K	T_∞	298.15 K
A	5.3×10^{-5} W/m·K^2	B	1.17×10^{-2} W/m·K

Appendix B. Dimensionless form

We choose characteristic length $L_c = R_0$ and time $T_c = t_d = 1/f_d$ to non-dimensionalize our model (21):

$$\ddot{a}_n^* + (A_n^* + A_{n_{rot}}^*)\dot{a}_n^* + (B_n^* + B_{n_{rot}}^*)a_n^* + C_{n_{rot}}^* = 0, \tag{B.1}$$

where

$$A_n^* = 3\frac{\dot{R}^*}{R^*} + \frac{2(n+1)(n+2)}{\mathrm{Re}R^{*2}}, \tag{B.2}$$

$$B_n^* = -(n-1)\frac{\ddot{R}^*}{R^*} + 4(n-1)(n+1)\frac{\dot{R}^*}{\mathrm{Re}\,R^{*3}} + \frac{(n-1)(n+1)(n+2)}{\mathrm{We}R^{*3}} + \frac{n+1}{\mathrm{Ca}R^{*2}}\left[2\frac{R_0}{L_c R^*} + 2\left(\frac{R_0}{L_c R^*}\right)^4 + \frac{n(n+1)}{1 + \frac{R_0}{L_c R^*} + \left(\frac{R_0}{L_c R^*}\right)^2}\right], \tag{B.3}$$

$$A_{n_{rot}}^* = -\frac{2n(n+1)(n+2)}{\mathrm{Re}R^{*2}}, \tag{B.4}$$

$$B^*_{n_{rot}} = 2n(n-1)(n+1)\frac{\dot{R}^*}{\mathrm{Re}R^{*3}} - \frac{n(n+1)^2}{\mathrm{Ca}R^{*2}}\left(\frac{R_0}{L_cR^*}\right)^4, \tag{B.5}$$

$$C^*_{n_{rot}} = \frac{n(n+1)(n+2)}{\mathrm{Re}R^{*2}}\mathscr{T}^*(1,t^*) - n(n+1)\frac{\dot{R}^*}{R^*}\int_1^\infty\left[1-\left(\frac{1}{y}\right)^3\right]\left(\frac{1}{y}\right)^n\mathscr{T}^*(y,t^*)dy, \tag{B.6}$$

where $\mathrm{Re} = \rho V_c L_c/\mu$ is the Reynolds number, $\mathrm{We} = \rho V_c^2 L_c/\sigma$ is the Weber number, $\mathrm{Ca} = \rho V_c^2/G$ is the Cauchy number, and $V_c = L_c/T_c$. The PDE for toroidal field (17) is

$$\frac{\partial\mathscr{T}^*}{\partial t^*} - \frac{\dot{R}^*}{R^*}\left(y-\frac{1}{y^2}\right)\frac{\partial\mathscr{T}^*}{\partial y} - \frac{1}{\mathrm{Re}R^{*2}}\frac{\partial^2\mathscr{T}^*}{\partial y^2} + \left[\frac{n(n+1)}{\mathrm{Re}R^{*2}}\frac{1}{y^2} - \frac{2\dot{R}^*}{R^*}\frac{1}{y^3}\right]\mathscr{T}^* = 0, \tag{B.7}$$

where the boundary conditions are $\mathscr{T}^* \to 0$ as $y \to \infty$ and shear stress continuity (18):

$$T^*(1,t^*) = \frac{2(n+2)}{n+1}\dot{a}_n^* - \frac{2(n-1)}{n+1}\frac{\dot{R}^*}{R^*}a_n^* + \frac{\mathrm{Re}}{\mathrm{Ca}}\left(\frac{R_0}{L_cR^*}\right)^4 a_n^* - 2\int_1^\infty\frac{\mathscr{T}^*(y,t^*)}{y^n}dy. \tag{B.8}$$

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